

Full Length Article

Multi-source geospatial fusion for enhanced visual geo-localization of unmanned aerial vehicles^{*}Xiong Qiu^{ID}^{*}, Shouyi Liao^{*}, Dongfang Yang, Ting Li, Shicheng Wang

Rocket Force University of Engineering, Xi'an, China

ARTICLE INFO

Keywords:

Unmanned aerial vehicles
Visual geo-localization
Satellite remote sensing imagery
Neural network
Feature extraction and matching

ABSTRACT

Currently, although significant advancements have been made in the autonomous navigation of Unmanned Aerial Vehicles (UAVs), reliable geo-localization remains a challenge when satellite navigation signals fail. Existing research primarily focuses on cross-view image retrieval, feature point matching, and visual localization based on three-dimensional (3D) reconstruction. While these methods work effectively in small-scale flight scenarios with pre-acquired image data, they struggle in larger-scale environments lacking prior information. To bridge this gap, we propose a visual geo-localization framework using multi-source geospatial fusion, which consists of two main phases: an offline phase for constructing multi-source databases and an online hierarchical geo-localization phase. During the offline phase, satellite imagery undergoes multi-perspective augmentation through rotational and translational transformations, followed by extraction of global and local features using neural networks to build feature databases. Simultaneously, terrain elevation data within the satellite imagery range is preserved for localization inference during the online phase. The online phase employs cascade matching: first, a coarse search based on global descriptors rapidly narrows potential locations; second, local feature matching obtains the pixel coordinates of matching feature points in aerial images and satellite images; third, geographic coordinates for the matched feature points are derived from satellite imagery and terrain elevation data, and the UAV geo-localization is achieved by leveraging stereoscopic projection geometry. We conducted experimental validation across geographically distinct test sites, covering an area of 26.9 km by 7.03 km, using aerial image captured at approximately 500 m altitude from both nadir view (90°) and oblique view (45° tilt). The results confirm the effectiveness of our proposed approach.

1. Introduction

Navigation, as a core technology for flight missions, directly determines flight safety and mission effectiveness in complex environments. The Global Positioning System (GPS) is the primary geospatial reference for UAVs, but is vulnerable to multipath interference and atmospheric anomalies, resulting in signal attenuation (Alkhatib et al., 2025; Ye et al., 2024). Therefore, it is crucial to adopt alternative localization methods under GPS denial conditions (Chen et al., 2024; Nafea et al., 2024). The expanding accessibility of multispectral satellite observation data has catalyzed advancements in image-based visual localization methods. Existing research predominantly concentrates on multi-view image retrieval and invariant feature extraction, yet image-based geo-localization systems remain underdeveloped. Some researchers (Chen et al., 2021; Lin et al., 2025) have utilized photogrammetric mapping prior to missions to build 3D terrain models for feature matching, thereby enabling UAV visual localization. Nevertheless, the computa-

tional overhead of dense reconstruction limits scalability, particularly in large flight areas requiring real-time processing, and this method is not applicable to unmapped areas that lack aerial data and cannot be pre-flown.

To address these critical limitations—including over-reliance on pre-built 3D terrain models, prohibitive computational costs for large-scale dense reconstruction, incompatibility with unmapped areas, and lack of support for non-strict nadir photography—we focus on UAV visual geo-localization in large scenes under low-availability conditions (i.e., GPS unavailability, no prior flight-area imagery, and infeasible pre-flight). This work aims to meet several requirements: supporting non-strict nadir aerial photography, enabling 3D (latitude, longitude, elevation) localization, and achieving geo-localization without prior knowledge of the UAV's launch point (including re-localization after prolonged failures). To achieve this, we propose a multi-source geospatial fusion method utilizing satellite remote sensing imagery and terrain elevation imagery to enhance the UAV geo-localization, as illustrated

^{*} This work was supported by National Natural Science Foundation of China 61673386 and 61603398.

^{*} Corresponding authors.

E-mail addresses: qxiong1991@163.com (X. Qiu), lsy_nudt_sec@sohu.com (S. Liao).

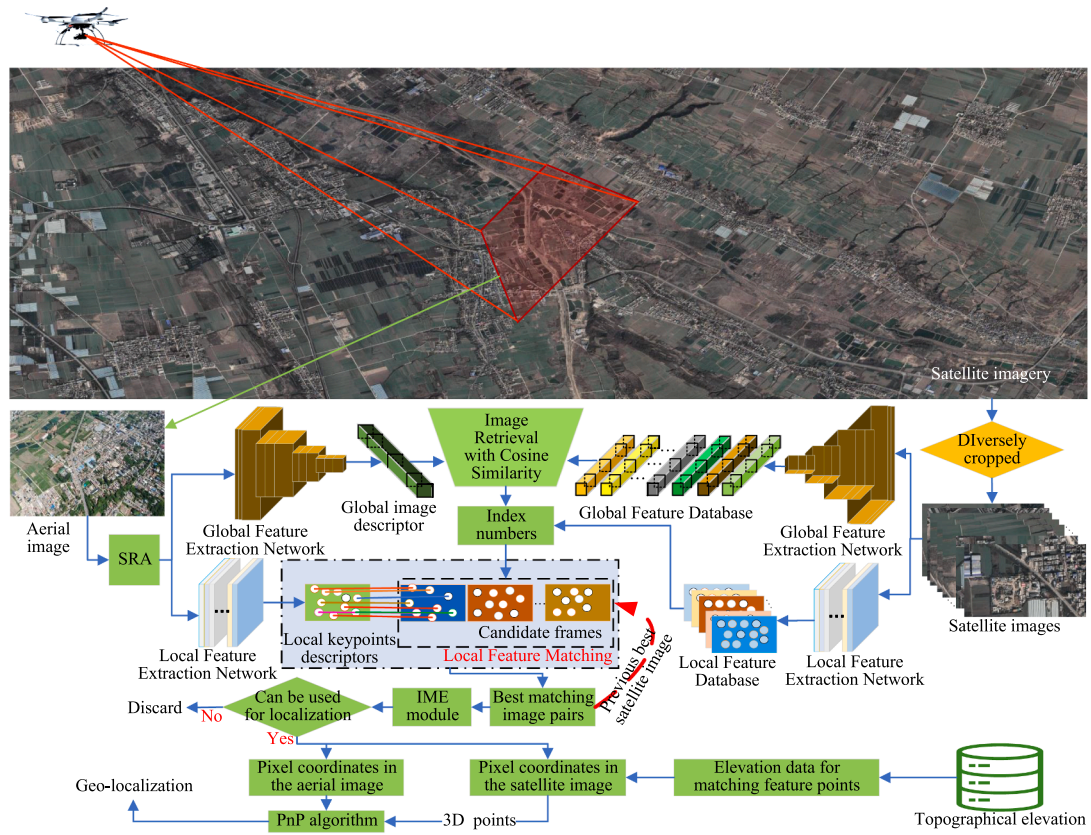


Fig. 1. Principles of the visual geo-localization using multi-source geospatial fusion.

in Fig. 1. This method consists of offline multi-source databases construction and online hierarchical geo-localization. Offline multi-source databases construction: Multi-view enhanced satellite images are generated by rotating and horizontal cropping methods, and deep neural networks are utilized to extract global features and local features to build feature databases. Additionally, we extract and store terrain elevation data for online localization inference. Online hierarchical geo-localization: First, the spatial resolution adaptive (SRA) method adjusts the pixel size of aerial images to align their spatial resolution with that of satellite images, effectively reducing feature matching bias caused by resolution differences. Then, aerial images are processed by two distinct feature extraction networks to generate global image descriptors and local keypoint descriptors. Using the global descriptor, a set of candidate satellite images similar to the aerial image is retrieved via cosine similarity. Meanwhile, the best matching satellite image obtained from the previous aerial image is added to the candidate set, leveraging spatial continuity between aerial images to reduce the risk of retrieval omissions. Subsequently, the invariant moment extraction (IME) module is built according to the local feature matching to determine whether the aerial images are suitable for localization. If suitable, the joint satellite imagery and terrain elevation data will provide geographic coordinates for the matched points, thus enabling visual geo-localization of the UAV. Otherwise, the image will be discarded. The main contributions of this work are summarized as follows:

- We propose a multi-source geospatial fusion framework for UAV visual geo-localization under low-availability conditions (GPS failure, lack of prior flight data, and unmappable areas), extending traditional 2D planar localization to 3D spatial localization.
- Our method is based on the principle of feature point stereo projection with terrain elevation prior constraints, making it adaptable to camera configurations at various tilt angles.

- Our method can achieve the UAV re-localization after a prolonged GPS failure, without requiring any prior information of the UAV's take-off location and flight path.
- We introduce a method integrating camera tilt angle, SRA, and IME to assess whether aerial images captured from a nadir view and an oblique view are suitable for localization.

The paper is structured as follow: Section 2 introduces related work on visual geo-localization; Section 3 presents our method, including of offline multi-source databases construction and online hierarchical geo-localization; Section 4 conducts experiments; Section 5 concludes the paper and discusses future research.

2. Background and related work

The issue of geo-localization is a classic research topic in the fields of computer vision and remote sensing. Early studies primarily utilized Digital Elevation Models (DEM) for terrain feature matching to achieve localization estimation based on terrain contours. These methods rely on high-precision terrain information, and acquiring high-quality DEM data typically depends on expensive remote sensing equipment and complex data processing workflows, which limits their application. Moreover, terrain contour-based matching methods perform poorly in flat areas due to the lack of distinctive terrain features. With the development of feature matching techniques, researchers have gradually shifted from manually designed image features to deep learning-based feature representation methods. These approaches can more effectively capture texture and geometric information in images, enabling cross-view image retrieval, feature matching, and achieving image-based visual localization. The comparison of key attributes of multiple advanced visual geo-localization methods is shown in Table 1.

Table 1
Comparison of key attributes of multiple advanced visual geo-localization methods.

Name	UAV viewpoint	Map size	Pre-flight preparation	Prior position	Absolute altitude
Luo et al. (2024)	arbitrary	1.2 km ²	Satellite imagery	No	No
Yao et al. (2024)	arbitrary	10 km ²	Satellite imagery + DEM	Yes	Yes
Chen et al. (2021)	arbitrary	400 × 400 m ²	3D model	No	Yes
Goforth and Lucey (2019)	top view	0.85 km ²	Satellite imagery	Yes	No
Wu et al. (2024)	arbitrary	2.5 km ²	3D model	No	Yes
Patel et al. (2020)	top view	300 × 350 m ²	3D model	Yes	Yes
Gurgu et al. (2022)	top view	–	Satellite imagery	No	No
Ours	arbitrary	196.4 km ²	Satellite imagery + DEM	No	Yes

2.1. Image-based localization

Image-based Localization has evolved significantly with advancements in local feature descriptors, enabling more compact and efficient representations of visual information. Contemporary methodologies in this field are broadly categorized into two paradigms: Relative Visual Localization (RVL) and Absolute Visual Localization (AVL) (Couturier & Akhloufi, 2021; Fan et al., 2025b). The main method of RVL is Visual Odometry (VO), which estimates a UAV's pose relative to its initial camera coordinate system through feature point matching between consecutive keyframes. While effective for short-term navigation, this approach requires prior positional data during initialization and suffers from error accumulation over extended trajectories. To mitigate this limitation, researchers have integrated inertial measurement unit (IMU) data with VO frameworks (Lynen et al., 2020; Si et al., 2023), leveraging sensor fusion techniques such as Kalman filtering to partially compensate for drift. Nevertheless, residual cumulative errors persist, particularly in large-scale or long-duration missions. In contrast, AVL (Li et al., 2025a; Yao et al., 2024) operates without dependency on initial positional priors by establishing pixel-level correspondences between aerial image and georeferenced maps. This direct alignment with georeferenced maps eliminates error propagation, making AVL particularly advantageous for high-precision, endurance-critical UAV applications. Recent developments (Aslan et al., 2022; Wang & Qi, 2024) in deep learning-enhanced feature matching and multi-sensor calibration have further enhanced AVL's robustness against environmental variations like lighting changes and seasonal landscape alterations.

2.2. Methods for AVL

Currently, there are mainly two visual geo-localization methods for AVL: one is based on 3D reconstruction, which requires a small flight area and prior flights; the other is based on satellite remote sensing imagery providing geographic information assistance, which is suitable for large flight scenarios and does not require 3D reconstruction of the flight scene through prior flights. Literature (Luo et al., 2024) employs Structure-from-Motion (SfM) to achieve 3D reconstruction, enabling three degrees of freedom (3-DOF) UAV localization by providing geographic coordinates for matched feature points. In contrast, Literature (Qiu et al., 2025) introduced a planar assumption-based localization framework that combines nadir-view aerial images with satellite imagery to enable 2-DOF geo-localization. As our aim is to realize visual geo-localization of the UAV in three-dimensional space under low assurance conditions where satellite navigation is denied and prior information and image data cannot be obtained through pre-flight, the visual geo-localization method using satellite remote sensing imagery has higher reference value for our task. Meanwhile, emerging cross-view matching techniques (Le Jeune et al., 2025) that align ground-level perspectives with aerial images also present promising extensions for our study.

2.3. Global image descriptors

Effectively capturing the comprehensive characteristics of an image is essential, and global image descriptors play a vital role in achieving this goal (Wang et al., 2024b; Yuan et al., 2022). Traditionally, global descriptors rely on handcrafted methods, including the Bag of Words (Goh et al., 2014), Fisher Vector (Gürünoğlu et al., 2022), and Vector of Locally Aggregated Descriptors (VLAD) (Faber et al., 2023). Among these, VLAD marked a significant advancement by aggregating local feature residuals to form compact global representations, resulting in improved retrieval precision compared to both the Bag of Words and the Fisher Vector on large-scale datasets. However, with the development of deep learning (Albdair et al., 2025; Song et al., 2025), Neural architectures like NetBoW (Liu et al., 2020) and NetVLAD (Khaliq et al., 2022) disrupted traditional paradigms through end-to-end learned representations. For example, MuSe-Net (Wang et al., 2024a) addresses the domain shift between training and testing environments by adaptively extracting environment-related features. Overall, neural network-based global descriptors benefit from large-scale end-to-end training, automatically learning optimal feature representations that capture higher-level semantics (Al-Qaysi et al., 2023; Yass & Faris, 2023), leading to superior performance and generalization compared to traditional methods.

2.4. Local keypoint descriptors

Local keypoint descriptors play a crucial role in encoding the visual content surrounding keypoints within an image. Early handcrafted descriptors, such as SIFT (Li et al., 2025b) and SURF (Wang et al., 2019), have historically achieved remarkable success in various image processing tasks. Despite their effectiveness, these traditional handcrafted descriptors have some limitations, including high computational costs, limited robustness to changes in viewpoint, and susceptibility to different seasonal and lighting conditions. To overcome these issues, neural network-based local keypoint descriptors have gradually emerged in recent years. Methods such as LF-Net (Li et al., 2024a), LoFTR (Liu et al., 2024), and SuperPoint (DeTone et al., 2018) can learn higher-level and more abstract feature representations through training on large-scale datasets, demonstrating improved generalization capabilities. Furthermore, advanced techniques like adaptive attention matching introduced in the literature (Fan et al., 2025a) can filter out useless information while acquiring more useful feature information, significantly improving the matching accuracy and efficiency of local features.

3. Methodology

To accurately determine the UAV's position, we established a world coordinate system based on the Mercator projection. The x -axis is aligned with the longitude direction, with its origin set at the Prime Meridian; the y -axis is aligned with the latitude direction, with its origin set at the Equator; the z -axis points vertically upward, following the right-hand rule. The reference ellipsoid is adopted as the datum surface to ensure that the positioning results obtained by the visual method share the same reference datum as the ellipsoidal height provided by satellite navigation. In the following section, we will detail

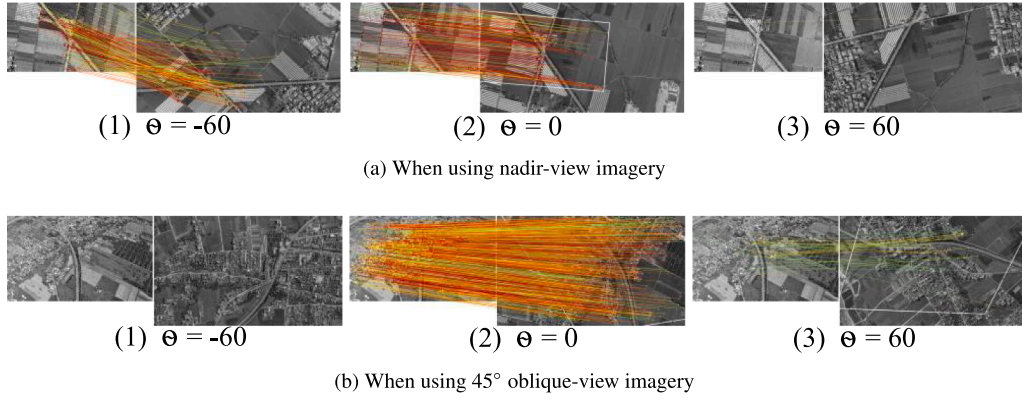


Fig. 2. When captured from various viewpoints, feature matching between aerial (left) and satellite images (right).

our proposed method, which encompasses both the offline multi-source databases construction and online hierarchical geo-localization.

3.1. Offline multi-source databases construction

To enhance localization efficiency during the online phase, we utilize feature descriptors from satellite images instead of high-resolution satellite imagery. This involves pre-cropping the satellite imagery to create feature databases. Additionally, terrain elevation data corresponding to the satellite image coverage area is stored to enhance spatial awareness and accuracy during localization.

3.1.1. A diverse cropping strategy to enhance satellite image matching

To verify the correlation between feature matching of aerial images and satellite images and the orientation of features, we use rectangular cropping frames at different azimuth angles to crop the corresponding satellite images. When aerial images are captured from various viewpoints, matching results with satellite images cropped with different azimuth angles is shown in Fig. 2, where θ represents cropping frame's rotation angle, and counterclockwise rotation is positive. When $\theta = 0$, the feature orientations of the two images are relatively aligned. The white box on the satellite image indicates the quadrilateral formed by projecting the aerial image's vertices. Fig. 2a and b show that robust feature matching occurs when the orientations of the two images are consistent, while significant differences in orientation can lead to significant degradation in matching performance. During the UAV flight, the camera captures images at various yaw angles, so it is essential to rotate the cropping frame at multiple angles to ensure that the cropped satellite image aligns directionally with the aerial image. Furthermore, to optimize the overlap between the aerial and satellite images, it is necessary to perform multi-variable translations of the cropping frame.

We utilize multi-variable translation (δ_W, δ_H) and rotation (θ) to crop satellite imagery, aligning the aspect ratios with aerial images, as illustrated in Fig. 3. The cropping workflow unfolds in distinct, visually guided steps: ① shows the original satellite imagery, which serves as the raw data for all cropping operations; ② shows direct cropping, where the orange grid indicates the cropping frame and the blue grid indicates the cropped images; ③ shows cropping with translation, highlighted by the yellow grid, where δ_W and δ_H specify the horizontal and vertical translation offsets, respectively. This translation step enriches the diversity of the images from ②; ④ shows cropping that incorporates both translation and rotation, marked by the red grid, where the yellow cropping frame is rotated counterclockwise by an angle θ . This approach increases the diversity of the cropped images, ensuring that aerial images captured at varying yaw angles maintain consistent feature orientations with satellite images, thus facilitating robust geo-localization. The cropping method is further detailed in three steps: ① indicates extracting the satellite image containing the pink grid; ② indicates clockwise

rotating this extracted image by an angle θ ; ③ indicates cropping the image based on pixel coordinates within the red frame. Each cropped satellite image is accompanied by key metadata, including translation amounts, height, width, rotation angle, and its index in the satellite imagery. This metadata enables subsequent calculations of 3D geographic coordinates of any feature points.

3.1.2. Global feature extraction from satellite images

Image preprocessing constitutes an indispensable step prior to feature extraction. The core objective of preprocessing lies in resizing satellite images to modify their spatial resolution, enabling individual pixels to capture richer texture features. This spatial modification improves the sensitivity of local features and enhances the representation of global image descriptors, which summarize the overall image characteristics. Spatial resolution of the original satellite image is denoted as RC_{sat} , with width W_{sat} and height H_{sat} . After preprocessing, the pixel sizes is W'_{sat} and H'_{sat} , the spatial resolution is denoted as RC'_{sat} , then,

$$\begin{cases} \frac{W_{sat}}{W'_{sat}} = \frac{H_{sat}}{H'_{sat}} \\ RC'_{sat} = RC_{sat} \cdot \frac{\max(W_{sat}, H_{sat})}{\max(W'_{sat}, H'_{sat})} \end{cases} \quad (1)$$

where $\max(M, N)$ denotes the greater between M and N . The global feature extraction framework we adopt, OpenIBL (a VGG + NetVLAD architecture), integrates the VGG (Visual Geometry Group) convolutional neural network, the NetVLAD feature aggregation mechanism, and a 2D convolutional layer for dimensionality reduction. Specifically, the VGG network converts input images into multi-channel feature maps with a feature dimension of 512; NetVLAD performs soft assignment of local features based on cluster centers, generating high-dimensional global image descriptors with an output feature dimension of $64 \times 512 = 32,768$, where 64 is the number of cluster centers and 512 is the original feature dimension; subsequently, the convolutional layer adjusts the shape and compresses the feature channels of this high-dimensional global image descriptor, effectively reducing the dimensionality from 32,768 to 4096. This framework is trained on the publicly available Pitts30K dataset (Zhang et al., 2024), and the tests were conducted on our flight dataset, which does not overlap in any way. Each satellite image generates a 4096-dimensional global feature vector through the global feature extraction framework, and the collection of these vectors from all satellite images constitutes the global feature database employed for aerial image retrieval in the online localization stage.

3.1.3. Local feature extraction from satellite images

We use SuperPoint (DeTone et al., 2018), a self-supervised deep learning architecture specifically designed for extracting highly distinctive local features. The SuperPoint network is trained through self-supervised learning, enabling it to automatically learn stable and

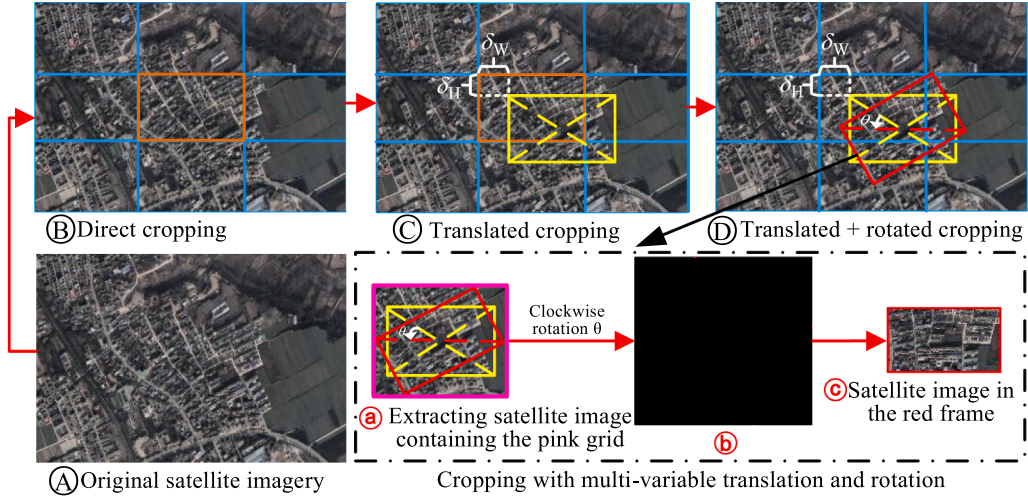


Fig. 3. The strategy for cropping satellite remote sensing imagery that includes rotation and translation variables.

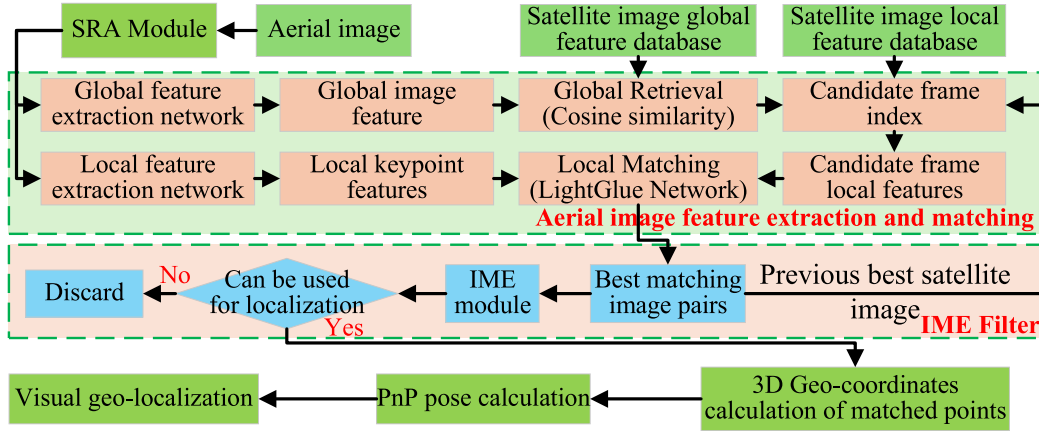


Fig. 4. Online hierarchical visual geo-localization workflow.

discriminative keypoints and their descriptors without the need for manual annotations. This network was trained on the MSCOCO dataset (Zhang et al., 2025), which contains 80,000 diverse images covering a wide range of scenes and object categories, significantly enhancing the model's generalization capability. Notably, there is no overlap between the MSCOCO dataset and the flight dataset we used, ensuring the model's adaptability and robustness in new environments. Using the trained SuperPoint model, we extracted local features from each satellite image, saving them individually in .npy file format. Collectively, these features from all satellite images constitute a comprehensive local feature database, which serves as the foundation for feature matching during the online localization process.

3.2. Online hierarchical geo-localization

The process of online hierarchical geo-localization involves several key steps: extracting and matching features from aerial images, evaluating the suitability of these images for localization, 3D coordinates estimation of matched feature points, and ultimately performing visual geo-localization. The online geo-localization workflow is illustrated in Fig. 4.

3.2.1. Extraction and matching of global and local features for aerial images

This part is divided into four main steps: Initially, the SRA module is employed to dynamically resize the aerial images, ensuring their spatial

resolution closely aligns with that of the satellite images. Next, the same pre-trained neural networks used in the offline phase (i.e., OpenIBL and SuperPoint) are applied to capture both global and local features from the aerial images. Following this, the satellite candidates are retrieved by cosine similarity of global descriptors, and the best-matched satellite image from the prior aerial frame is further added to form the complete candidate set. Finally, precise feature matching between the chosen candidate images and the aerial image are established using the neural network LightGlue (Lindemberger et al., 2023). The training of LightGlue involves two stages: initially, pre-training on synthetic homographies generated from 1 million images in the Oxford-Paris retrieval dataset; subsequently, fine-tuning on the MegaDepth dataset (Wilks et al., 2021). Notably, these two publicly available datasets do not overlap with our flight dataset.

3.2.2. Spatial resolution adaptive and invariant moments extraction

To address the impact of resolution differences on feature representation and matching performance, as well as the challenge of evaluating feature matching validity, we introduce two deterministic, training-free processing modules: the Spatial Resolution Adaptive (SRA) module and the Invariant Moment Extraction (IME) module. The SRA module dynamically adjusts the pixel dimensions of aerial images in real time, ensuring that aerial and satellite images achieve nearly identical spatial resolutions. This alignment effectively mitigates discrepancies in feature matching caused by resolution variations. Additionally, the IME module enhances the reliability of feature matching by assessing the geometric

consistency of aerial images before and after projection, filtering out low-quality frames with severe perspective distortion that cannot support reliable matching.

Traditionally, the usability of an aerial image for localization is determined by whether the number of matched feature points exceeds a predefined threshold and if the projected geometric center of the aerial image lies within the satellite image boundaries. Formally, the normalized coordinates of the projected center should satisfy,

$$\begin{cases} 0 \leq \text{Normal}(\hat{x}'_c) = \frac{\hat{x}'_c}{W'_{\text{sat}}} \leq 1 \\ 0 \leq \text{Normal}(\hat{y}'_c) = \frac{\hat{y}'_c}{H'_{\text{sat}}} \leq 1 \end{cases} \quad (2)$$

where $(\hat{x}'_c, \hat{y}'_c)$ denotes the pixel coordinate of the aerial image's geometric center projected onto the satellite image. $(\text{Normal}(\hat{x}'_c), \text{Normal}(\hat{y}'_c))$ denotes the normalized coordinates of $(\hat{x}'_c, \hat{y}'_c)$. When aerial images have a fixed size, variations in flight altitude, camera tilt angle, and focal length lead to significant differences in spatial resolution. The spatial resolution RC_{uav} before online processing can be expressed as,

$$\frac{\Delta \cdot f_c}{\Delta \cdot f_c + h / \cos \tau} = \frac{\Delta \cdot W_{\text{uav}}}{\text{RC}_{\text{uav}} \cdot W_{\text{uav}}} \quad (3)$$

$$\text{RC}_{\text{uav}} = \Delta + \frac{h}{f_c \cdot \cos \tau} \approx \frac{h}{f_c \cdot \cos \tau} \quad (4)$$

where h is the UAV's relative altitude above the ground area captured in the aerial image, determined through the visual method; Δ denotes the camera pixel pitch; W_{uav} is the width of the aerial image; f_c is the camera's calibrated focal length in pixels; and τ is the camera tilt angle, defined as the angle between the camera's optical axis and the gravity direction, estimated by Eq. (19). When processing an aerial image online at a new spatial resolution RC'_{uav} , with width W'_{uav} and height H'_{uav} , the SRA adjustment is formulated as,

$$\begin{cases} \text{RC}'_{\text{uav}} \approx \cos(\text{Lat}_V) \cdot \text{RC}_{\text{sat}} \\ \max(H'_{\text{uav}}, W'_{\text{uav}}) = \frac{\text{RC}_{\text{uav}} \cdot \max(H_{\text{uav}}, W_{\text{uav}})}{\text{RC}'_{\text{uav}}} \end{cases} \quad (5)$$

where Lat_V denotes the average latitude of matched feature points. After adaptive adjustment of aerial image sizes through SRA, the feature matching results between aerial images and satellite images, as well as the reprojection effects of aerial images onto satellite images, are shown in Fig. 5. "Reprojection" depicts the aerial images projected onto the satellite images, where the darker background layer is the satellite image and the overlaid lighter region indicates the projected aerial coverage. Fig. 5a illustrates the case where the camera adopts a nadir view. The feature matching shows a good parallel relationship, and the projection of the aerial image onto the satellite image involves only translation, rotation, and slight scaling changes. Fig. 5b shows that when the camera adopts an oblique view, due to perspective effects, the spatial resolution of the lower part of the aerial image near the camera is relatively low, while the spatial resolution of the upper part farther from the camera is relatively high. This causes non-uniform scaling in the projection of different regions of the aerial image onto the satellite image: the upper projection area of the aerial image is larger, the lower projection area is smaller, and the farther from the center of the aerial image, the more significant the scaling differences.

To address this issue, we designed the IME module, which has rotation, translation, and scaling invariance features, to determine whether the aerial image is suitable for visual localization. Treating all feature points uniformly, the spatial moments M_{pq} can be expressed as,

$$\begin{cases} M_{pq} = \sum_{x,y} (\text{array}(x, y) \cdot x^p \cdot y^q) \\ \bar{x} = \frac{M_{10}}{M_{00}}, \quad \bar{y} = \frac{M_{01}}{M_{00}} \end{cases} \quad (6)$$

where $\text{array}(x, y)$ denotes the gray value at pixel coordinate (x, y) , which takes the value of 1; $(\bar{x}, \bar{y})$ denotes the pixel coordinates of the image's geometric center. At this point, The central moment μ_{pq} and normalized central moment μ_{pq} can be expressed as,

$$\begin{cases} \mu_{pq} = \sum_{x,y} (\text{array}(x, y) \cdot (x - \bar{x})^p \cdot (y - \bar{y})^q) \\ \mu_{pq} = \frac{\mu_{pq}}{\mu_{00}^{(p+q)/2+1}} \end{cases} \quad (7)$$

where the center moments have translational invariance, while the normalized center moments have both translational and scalability invariance. According to reference (Qiu et al., 2024), seven image invariant moments derived from normalized center moments demonstrate invariance to rotation, translation, and scaling, which can be expressed as,

$$\begin{cases} h_0 = \mu_{20} + \mu_{02} \\ h_1 = (\mu_{20} - \mu_{02})^2 + 4\mu_{11}^2 \\ h_2 = (\mu_{30} - 3\mu_{12})^2 + (3\mu_{21} - \mu_{03})^2 \\ h_3 = (\mu_{30} + \mu_{12})^2 + (\mu_{21} + \mu_{03})^2 \\ h_4 = (\mu_{30} - 3\mu_{12})(\mu_{30} + \mu_{12})[(\mu_{30} + \mu_{12})^2 - 3(\mu_{21} + \mu_{03})^2] + (3\mu_{21} - \mu_{03})(\mu_{21} + \mu_{03})[3(\mu_{30} + \mu_{12})^2 - (\mu_{21} + \mu_{03})^2] \\ h_5 = (\mu_{30} - \mu_{02})[(\mu_{30} + \mu_{12})^2 - (\mu_{21} + \mu_{03})^2] + 4\mu_{11}(\mu_{30} + \mu_{12})(\mu_{21} + \mu_{03}) \\ h_6 = (3\mu_{21} - \mu_{03})(\mu_{30} + \mu_{12})[(\mu_{30} + \mu_{12})^2 - 3(\mu_{21} + \mu_{03})^2] - (\mu_{30} - 3\mu_{12})(\mu_{21} + \mu_{03})[3(\mu_{30} + \mu_{12})^2 - (\mu_{21} + \mu_{03})^2] \end{cases} \quad (8)$$

Treating aerial images as uniform gray rectangular planes, $h_k = 0$ for $k = 2, 3, 4, 5, 6$, so we only focus on the relative errors of h_0 and h_1 , then,

$$\delta(h_k) = \frac{h'_k - h_k}{h_k}, \quad k = (0, 1) \quad (9)$$

where h_k and h'_k represent k th moment before and after projection, respectively, while δ_{h_k} denotes the relative error of this moment h_k . By establishing a permissible range for δ_{h_k} , we can filter out aerial images that only exhibit translation, rotation, and scaling transformations. These images tend to exhibit strong local feature matching with satellite images, making them suitable for the UAV localization. The relative error of the invariant moment reflects the degree of distortion other than translation, rotation, and scaling, which is related to the camera tilt angle. When the camera tilt angle is closer to 0, the variation in the invariant moments before and after projection becomes smaller.

Under the conditions of using the SRA module, when the camera captures aerial images with a nadir view, the projection onto the satellite image involves negligible scaling. Thus, we eliminate aerial images with a larger scaling by leveraging the relative error of the area. The relative error δ_{area} can be expressed as,

$$\delta_{\text{area}} = \frac{S'_{\text{UAV}} - S_{\text{UAV}}}{S_{\text{UAV}}} = \frac{M'_{00} - M_{00}}{M_{00}} \quad (10)$$

where S_{UAV} denotes the area before the aerial image projection, equivalent to the moment M_{00} ; while S'_{UAV} denotes the projected area of the aerial image, equivalent to the moment M'_{00} .

Aerial images are deemed suitable for localization if: (1) $\delta_{h_0}, \delta_{h_1} \in [-0.2, 0.2]$ and $\delta_{\text{area}} \in [-0.3, 1]$, with $\tau \in [-5^\circ, 5^\circ]$ (corresponding to near-nadir view), or (2) $\delta_{h_0}, \delta_{h_1} \notin [-0.1, 0.1]$, but still within $[-4.5, 4.5]$, with $\tau \notin [-5^\circ, 5^\circ]$ (corresponding to oblique view). This differentiated threshold design fully adapts to the severe perspective distortion of oblique imagery and avoids incorrect elimination of valid frames caused by unified and overly strict constraints. It is important to emphasize that when aerial and satellite images have reliable feature matching, $\delta_{h_0}, \delta_{h_1} \in [-2, 2]$. The broader range of $\delta_{h_0}, \delta_{h_1} \in [-4.5, 4.5]$ is specifically designed to accommodate projection deformation differences under extreme terrain conditions, ensuring sufficient adaptability in various unknown environments and effectively reducing parameter dependence on specific datasets. Conversely, when the feature matching is poor, δ_{h_0} and δ_{h_1} tend to increase significantly, often reaching hundreds or even higher.

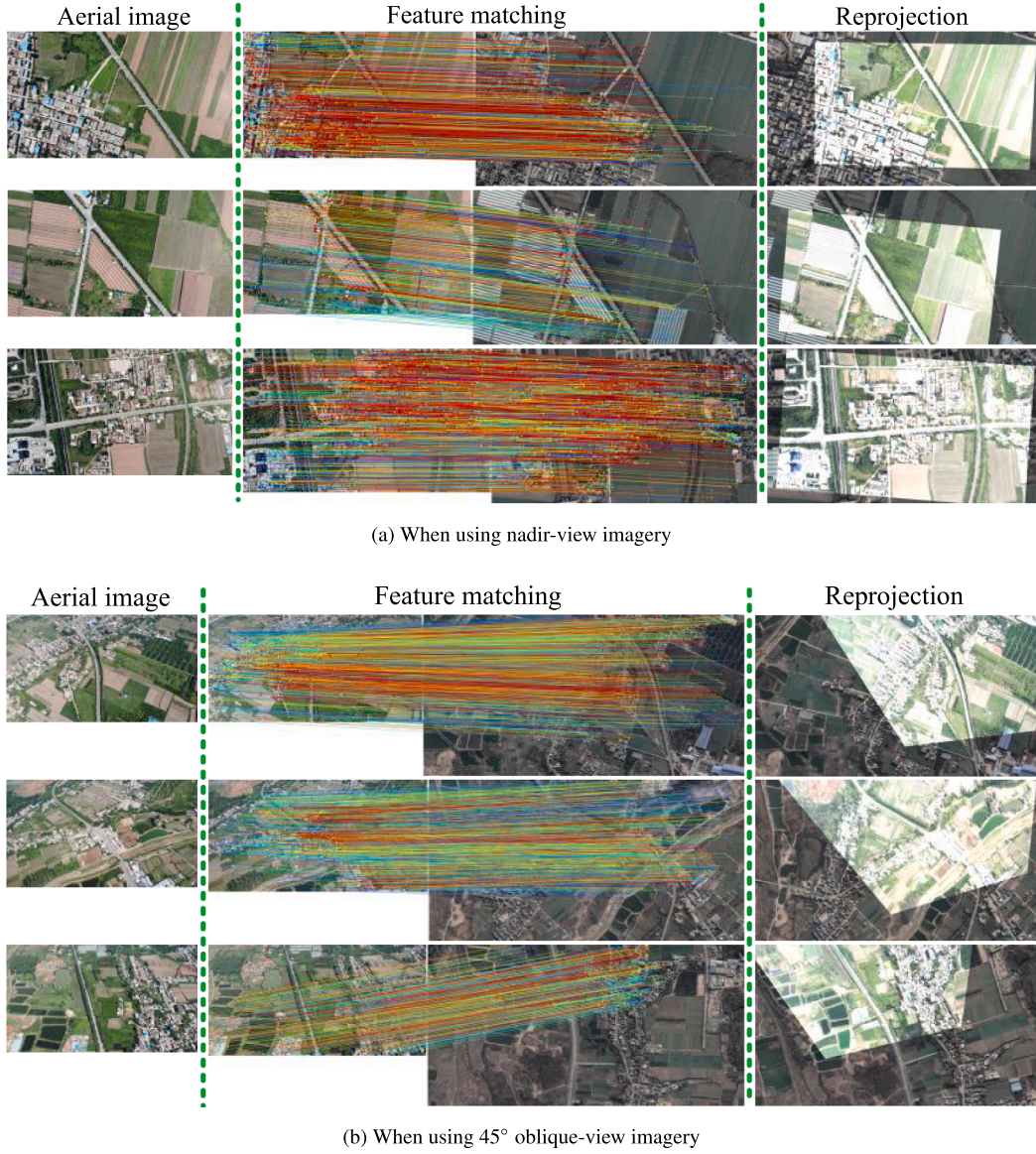


Fig. 5. Feature matching between aerial images and satellite images and reprojection of aerial images onto satellite images.

3.2.3. 3D coordinates estimation of matched feature points

Assuming the column and row indices of the satellite image in satellite imagery is i and j , respectively. The cropping frame undergoes translations δ_W and δ_H along the width and height directions, and a rotation by angle θ . For the l th feature point located on the satellite image, its original coordinates are given by $(u(i, j, l), v(i, j, l))$, then the coordinates $(u'(i, j, l), v'(i, j, l))$ of the feature point in the satellite imagery can be expressed as,

$$\begin{cases} u'(i, j, l) = \delta_W + i \cdot W_{\text{sat}} + \cos \theta \cdot \left(u(i, j, l) - \frac{W_{\text{sat}}}{2} \right) \\ \quad - \sin \theta \cdot \left(v(i, j, l) - \frac{H_{\text{sat}}}{2} \right) + \frac{W_{\text{sat}}}{2} \\ v'(i, j, l) = \delta_H + j \cdot H_{\text{sat}} + \sin \theta \cdot \left(u(i, j, l) - \frac{W_{\text{sat}}}{2} \right) \\ \quad + \cos \theta \cdot \left(v(i, j, l) - \frac{H_{\text{sat}}}{2} \right) + \frac{H_{\text{sat}}}{2} \end{cases} \quad (11)$$

According to the affine transformation principle, convert the pixel coordinates $(u'(i, j, l), v'(i, j, l))$ to the coordinates $(x_M(i, j, l), y_M(i, j, l))$

on the Mercator projection plane, then

$$\begin{bmatrix} x_M(i, j, l) \\ y_M(i, j, l) \end{bmatrix} = \begin{bmatrix} A_{PM} & B_{PM} \\ D_{PM} & E_{PM} \end{bmatrix} \begin{bmatrix} u'(i, j, l) \\ v'(i, j, l) \end{bmatrix} + \begin{bmatrix} C_{PM} \\ F_{PM} \end{bmatrix} \quad (12)$$

where $A_{PM}, B_{PM}, C_{PM}, D_{PM}, E_{PM}$ and F_{PM} represent the affine transformation parameters for converting pixel coordinates to Mercator projection plane. Based on the pixel coordinates of four vertices in the satellite imagery and their corresponding Mercator coordinates, the calculation of the affine transformation parameters can be described as,

$$\begin{bmatrix} x_M(\text{tl}) \\ x_M(\text{tr}) \\ x_M(\text{br}) \\ x_M(\text{bl}) \\ y_M(\text{tl}) \\ y_M(\text{tr}) \\ y_M(\text{br}) \\ y_M(\text{bl}) \end{bmatrix} = \begin{bmatrix} u(\text{tl}) & v(\text{tl}) & 1 & 0 & 0 & 0 \\ u(\text{tr}) & v(\text{tr}) & 1 & 0 & 0 & 0 \\ u(\text{br}) & v(\text{br}) & 1 & 0 & 0 & 0 \\ u(\text{bl}) & v(\text{bl}) & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & u(\text{tl}) & v(\text{tl}) & 1 \\ 0 & 0 & 0 & u(\text{tr}) & v(\text{tr}) & 1 \\ 0 & 0 & 0 & u(\text{br}) & v(\text{br}) & 1 \\ 0 & 0 & 0 & u(\text{bl}) & v(\text{bl}) & 1 \end{bmatrix} \begin{bmatrix} A_{PM} \\ B_{PM} \\ C_{PM} \\ D_{PM} \\ E_{PM} \\ F_{PM} \end{bmatrix} \Rightarrow \mathbf{V}_x = \mathbf{M} \cdot \mathbf{S}$$

(13)

where the vector $\mathbf{V}_X$ is the collection of Mercator coordinates of the four vertices, the matrix $\mathbf{M}$ is formed by the pixel coordinates of the four vertices, and the vector $\mathbf{S}$ contains the affine transformation parameters. The affine transformation parameters are solved using the least squares method.

Convert the Mercator coordinates $(x_M(i, j, l), y_M(i, j, l))$ of the feature points to the pixel coordinates $(u''(i, j, l), v''(i, j, l))$ in the terrain elevation, then

$$\begin{bmatrix} u''(i, j, l) \\ v''(i, j, l) \end{bmatrix} = \text{Round} \left(\begin{bmatrix} A_{MP} & B_{MP} \\ D_{MP} & E_{MP} \end{bmatrix} \cdot \begin{bmatrix} x_M(i, j, l) \\ y_M(i, j, l) \end{bmatrix} + \begin{bmatrix} C_{MP} \\ F_{MP} \end{bmatrix} \right) \quad (14)$$

where A_{MP} , B_{MP} , C_{MP} , D_{MP} , E_{MP} , and F_{MP} are the affine transformation parameters for converting Mercator coordinates to pixel coordinates in the terrain elevation imagery. These parameters are obtained by solving a linear system analogous to Eq. (13), constructed using the pixel coordinates of the four vertices in the terrain elevation imagery and their corresponding Mercator coordinates. $\text{Round}(\mathbf{U})$ indicates rounding each element in the vector $\mathbf{U}$. Then, the coordinates $(x_W(i, j, l), y_W(i, j, l), z_W(i, j, l))$ of the matching feature point in the world coordinate system can be expressed as,

$$\begin{cases} x_W(i, j, l) = x_M(i, j, l) \\ y_W(i, j, l) = y_M(i, j, l) \\ z_W(i, j, l) = \frac{\mathbf{HD}(u''(i, j, l), v''(i, j, l)) + N(\text{Lat}(i, j, l))}{\cos(\text{Lat}(i, j, l))} \\ N(\text{Lat}(i, j, l)) = \frac{a}{\sqrt{1 - e^2 \sin^2 \text{Lat}(i, j, l)}} \\ e^2 = \frac{a^2 - b^2}{a^2} \end{cases} \quad (15)$$

where $\mathbf{HD}(u, v)$ denotes the feature point elevation database stored based on pixel coordinates (u, v) , $N(\text{Lat}(i, j, l))$ is the distance between the reference ellipsoid and the geoid; a stands for the Earth's semi-major axis, valued at 6,378,137 m, and b represents the Earth's semi-minor axis, valued at 6,356,752 m; $\text{Lat}(i, j, l)$ denotes the latitude of the matching point obtained through visual methods.

3.2.4. Visual geo-localization

When the coordinates of feature points in the world coordinate system and the pixel coordinates of these points after stereo projection onto aerial images are known, the mapping relationship between 3D spatial points and 2D image projection points can be established using the camera's intrinsic and extrinsic parameters, thereby achieving the estimation of the UAV's position. This process can be expressed as,

$$s_d \cdot \begin{bmatrix} u_d \\ v_d \\ 1 \end{bmatrix} = \mathbf{K}_C \cdot \left(\mathbf{R}_W^C \cdot \begin{bmatrix} x_W(i, j, l) \\ y_W(i, j, l) \\ z_W(i, j, l) \end{bmatrix} + \mathbf{t}_{CW} \right) \quad (16)$$

where s_d denotes the scaling parameter; $\mathbf{R}_W^C$ represents the transformation from the world coordinate system to the camera coordinate system; $\mathbf{t}_{CW}$ indicates the position of the origin of the world coordinate system expressed in the camera coordinate system; and $\mathbf{K}_C$ represents the camera intrinsic parameter matrix. We employ OpenCV's built-in PnP solver along with the RANSAC algorithm to reject outliers to solve for $\mathbf{R}_W^C$ and $\mathbf{t}_{CW}$. However, the camera's attitude angle is related to the rotation matrix $\mathbf{R}_{C_0}^C$, which transforms coordinates from the camera's initial frame to its current frame. Meanwhile, the camera's position is the translation vector $\mathbf{t}_{WC}$. The matrix $\mathbf{R}_{C_0}^C$ and the vector $\mathbf{t}_{WC}$ can be described as,

$$\begin{cases} \mathbf{t}_{WC} = -(\mathbf{R}_W^C)^{-1} \cdot \mathbf{t}_{CW} \\ \mathbf{R}_{C_0}^C = \mathbf{R}_W^C \mathbf{R}_{C_0}^W \end{cases} \quad (17)$$

where $\mathbf{t}_{WC}$ represents the camera's position in the world coordinate system, corresponding to the UAV's geo-localization; $(\mathbf{R}_W^C)^{-1}$ is the inverse of $\mathbf{R}_W^C$.

Since the IME module can eliminate aerial images with poor feature matching by combining the camera tilt angle, it is essential to convert $\mathbf{R}_{C_0}^C$ into Euler angles that represent the camera attitude. The Euler angles depend on the definition of the camera coordinate system and the sequence of rotations around each axis. In this paper, the camera's initial coordinate system is defined with the x -axis, y -axis, and z -axis representing the east, down, and north directions, respectively, with the camera's optical axis aligned along the z -axis. Rotations about the x -axis, y -axis and z -axis corresponds to pitch angle ϕ , yaw angle ψ and roll angle γ , respectively. Additionally, the rotation sequence is first around the y -axis, then the z -axis, and finally the x -axis. Thus, the rotation matrix $\mathbf{R}_{C_0}^C$ from the initial to current camera frame is,

$$\begin{aligned} \mathbf{R}_{C_0}^C &= \mathbf{R}_x(\phi) \cdot \mathbf{R}_z(\gamma) \cdot \mathbf{R}_y(\psi) \\ &= \begin{bmatrix} \cos \psi \cos \gamma & \sin \gamma & -\sin \psi \cos \gamma \\ -\cos \phi \cos \psi \sin \gamma + \sin \phi \sin \psi & \cos \phi \cos \gamma & \cos \phi \sin \psi \sin \gamma + \sin \phi \cos \psi \\ \sin \phi \cos \psi \sin \gamma + \cos \phi \sin \psi & -\sin \phi \cos \gamma & -\sin \phi \sin \psi \sin \gamma + \cos \phi \cos \psi \end{bmatrix} \end{aligned} \quad (18)$$

Since the gimbal system controls the roll angle γ to be 0, the camera's pitch and tilt angles can be represented as,

$$\begin{cases} \varphi_{\text{camera}} = -\arctan \left(\frac{\mathbf{R}_{C_0}^C(3, 2)}{\mathbf{R}_{C_0}^C(2, 2)} \right) \\ \tau = 90^\circ + \varphi_{\text{camera}} \end{cases} \quad (19)$$

where τ denotes the camera's tilt angle.

4. Experiment

4.1. Experimental details

We conducted two non-intersecting flight experiments in a large area to evaluate our proposed method. The flight area had widths of 26.9 km and heights of 7.03 km. The UAV was equipped with a gimbal-mounted optical camera capturing aerial images at 30 frames per second, with a resolution of 1920×1080 pixels. In the first flight, the UAV flew at a stable height of 500 m relative to the take-off point, with an absolute altitude of about 1064 m, capturing images from a nadir view. Additionally, aerial images collected during take-off and landing phases were included, covering a relative altitude range from 265 m to 500 m. In the second flight, the UAV flew at a stable height of 500 m relative to the take-off point, with an absolute altitude of about 970 m, capturing images at a 45° tilt angle. During the second flight, the UAV encountered two periods of GPS signal loss, so we used the aerial images from the phases of continuous GPS stability. Part of the aerial images taken by the gimbal camera and the satellite imagery used to build the offline database are illustrated in Fig. 6. Fig. 6a shows the partial aerial images of the first flight, Fig. 6b illustrates the partial aerial images of the second flight, and Fig. 6c shows the satellite remote sensing imagery, which is utilized to construct global and local feature databases. Our code is running in an environment with a graphics card of NVIDIA RTX3090, a CPU of Intel(R) Xeon(R) CPU E5-2660 v3, and PyTorch version 1.12.1. The localization time for each aerial image is approximately 2.1 s, including about 0.57 s for global retrieval, approximately 1.38 s for local feature matching, around 0.05 s for IME, and about 0.1 s for PnP pose estimation.

The satellite remote sensing imagery has a spatial resolution of 0.597 m/pixel in Mercator projection plane. To facilitate UAV visual localization at arbitrary yaw angles, the cropping frame was rotated in 20-degree increments and translated by $0.2W_{\text{sat}}$ and $0.2H_{\text{sat}}$, totaling 18 angles and 5 translations. The dimensions of the cropping frame were defined based on the geographic range that the satellite image should cover the aerial image. Therefore, at location 1, the cropped satellite image measures 1350×760 , resulting in a total of 39,035 images, while at



(a) At location 1



(b) At location 2



(c) Satellite remote sensing imagery (26.9 km in width, 7.03 km in height)

Fig. 6. Partial aerial images and satellite imagery used to build the offline database.

location 2, the cropped size is 2350×1322 , resulting in a total of 11,132 images. Additionally, the terrain elevation data extracted from the elevation imagery are saved in an npy file for real-time access during the online localization phase.

4.2. Evaluation method

To evaluate the UAV's localization accuracy, we define a local geographic coordinate system aligned with the world coordinate system, sharing the same origin and orientation. Unlike the world coordinate system, distances in this geographic system correspond to actual spatial measurements and are adjusted by the cosine of the local latitude. The Real Time Kinematic (RTK) localization is used as the ground truth, while the localization obtained through the visual method is considered the measured value. When the camera captures an aerial image, the localization error along the x , y , and z axes in the geographic coordinate system can be expressed as,

$$\begin{cases} \text{Error_x}(k) = x_R(k) \cdot \cos(\text{Lat}_R(k)) - x_V(k) \cdot \cos(\text{Lat}_V(k)) \\ \text{Error_y}(k) = y_R(k) \cdot \cos(\text{Lat}_R(k)) - y_V(k) \cdot \cos(\text{Lat}_V(k)) \\ \text{Error_z}(k) = z_R(k) - z_V(k) \cdot \cos(\text{Lat}_V(k)) \end{cases} \quad (20)$$

When the UAV takes the k th image, $x_R(k)$ and $y_R(k)$ indicate the UAV's position in the Mercator projection plane, derived from latitude and longitude measurements obtained by RTK; $z_R(k)$ indicates the UAV's ellipsoidal height relative to the reference ellipsoid, also measured by RTK; $\text{Lat}_R(k)$ indicates the UAV latitude measured by RTK; $x_V(k)$, $y_V(k)$, and

$z_V(k)$ indicate the UAV's localization obtained by our proposed method; $\text{Lat}_V(k)$ is the UAV's latitude derived by converting $y_V(k)$; $\text{Error_x}(k)$, $\text{Error_y}(k)$, and $\text{Error_z}(k)$ denote the UAV's localization errors along each axis within the geographic coordinate system.

We assess the UAV's localization accuracy throughout the flight by calculating the mean absolute error (MAE) and root mean square error (RMSE) within the geographic coordinate system. These metrics are evaluated along the x , y , and z axes, as well as combined in the xy plane and the full xyz space. Specifically, the mean absolute error MAE_xyz and root mean square error RMSE_xyz of the UAV in the xyz direction of the geographic coordinate system are represented as,

$$\begin{cases} \text{MAE_xyz} = \frac{\sum_{k=1}^N \sqrt{\text{Error_x}(k)^2 + \text{Error_y}(k)^2 + \text{Error_z}(k)^2}}{N} \\ \text{RMSE_xyz} = \sqrt{\frac{\sum_{k=1}^N (\text{Error_x}(k)^2 + \text{Error_y}(k)^2 + \text{Error_z}(k)^2)}{N}} \end{cases} \quad (21)$$

where N represents the number of aerial images available for localization.

4.3. Experimental analysis

To demonstrate the advantages of the combined use of the spatial resolution adaptive (SRA) method and invariant moment extraction (IME)

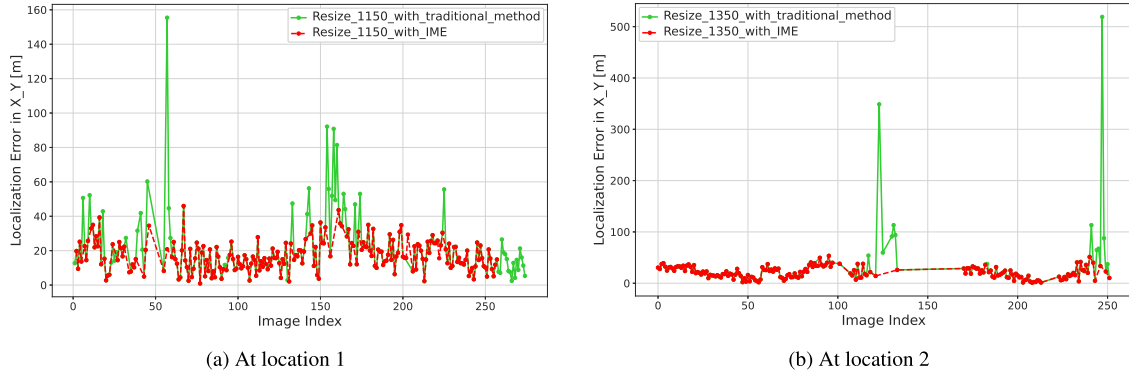


Fig. 7. Comparing the effects of the traditional method and the IME on the UAV localization errors at location 1 and 2.

Table 2

Comparing the numerical effects of the traditional method and the IME on the UAV localization errors at location 1 and 2.

Location	Frames	Method	Recall (%)	Localization MAE[m]			Localization RMSE[m]		
				x	y	xy	x	y	xy
location 1	275	Traditional	93.09	11.35	14.29	20.31	16.54	20.10	26.03
		IME	77.45	9.48	12.11	17.15	11.96	15.05	19.22
location 2	274	Traditional	68.98	14.41	21.71	28.33	28.03	46.56	54.34
		IME	64.96	10.80	15.73	20.98	15.10	18.42	23.82

module, as well as the advantages of our proposed multi-source geospatial fusion visual localization, we conducted five experiments.

4.3.1. Experiment 1

To assess the advantages of the IME over traditional methods for the selection of aerial images for localization, we conducted a comparison using both approaches. The widths of the aerial images at locations 1 and 2 were fixed at 1150 and 1350 pixels, respectively. The parameter setting of the traditional method: the threshold for matching feature points is set to 80, $0 \leq \text{Normal}(\hat{x}'_c) \leq 1$, and $0 \leq \text{Normal}(\hat{y}'_c) \leq 1$. Parameters of IME module: when $-0.2 \leq \delta_{h_0} \leq 0.2$, $-0.2 \leq \delta_{h_1} \leq 0.2$, and $-0.3 \leq \delta_{\text{area}} \leq 1$, the aerial image is considered suitable for localization if the tilt angle $\tau \in [-5^\circ, 5^\circ]$. When δ_{h_0} and δ_{h_1} are not within the range of $[-0.1, 0.1]$, but are within the range of $[-4.5, 4.5]$, and if $\tau \notin [-5^\circ, 5^\circ]$, the aerial image can still be considered suitable for localization. In all other cases, it is deemed unsuitable. When relying solely on satellite remote sensing imagery for localization, the UAV localization errors at locations 1 and 2 are shown in the Fig. 7, with numerical error data presented in Table 2. The horizontal axis indicates the sequence number of the aerial images, while the vertical axis indicates the comprehensive localization error in the geographic coordinate system along the x and y axes. The blue line corresponds to the traditional method, and the red line corresponds to the IME method.

As shown in Fig. 7a and b, when the aerial image pixel size is pre-fixed, our proposed IME module effectively eliminates larger localization errors compared to the traditional method. However, at location 1, compared to the traditional method, our proposed IME module rendered some aerial images that could have been used for localization during the descent phase unusable. The reason for this issue is that when there is a significant change in flight altitude, the pre-fixed size of the aerial images leads to a large discrepancy in spatial resolution compared to satellite images, resulting in excessive scaling before and after the projection of the aerial images, which causes the relative error of the area in the IME module to eliminate this portion of the aerial images. In contrast, at location 2, since the UAV's flight altitude remains relatively stable, the aerial images captured along the flight path do not suffer from becoming unusable for localization during the initial and final phases when using the IME module. A deeper analysis of Table 2 reveals that the IME module has effectively improved the UAV's localization accuracy com-

pared to traditional method, reducing the comprehensive localization error along the x and y axes from 20.31 m to 17.15 m at location 1, and from 28.33 m to 20.98 m at location 2. However, when the aerial image size is pre-fixed, reliable localization cannot be realized by relying only on the IME module at moments of large altitude changes such as takeoff and landing. This issue can be addressed by the SRA module in Experiment 2.

4.3.2. Experiment 2

This experiment aims to demonstrate that the combination of SRA and IME yields better results compared to using IME alone. When SRA is not used, the widths of the aerial images at location 1 were adjusted sequentially to 750, 950, 1150, and 1350 pixels; and at location 2, the widths were adjusted sequentially to 950, 1150, 1350, and 1550 pixels, and the IME module's parameter settings are consistent with those of Experiment 1. Using only satellite imagery for visual localization, we compare the localization errors of using the IME module alone and the combined SRA and IME modules. The UAV localization errors at both locations are shown in Fig. 8, and the quantitative results are shown in the Table 3. The blue and green lines indicates the localization results with pre-fixed aerial image pixel sizes when only the IME is used, and the red line indicates the localization results when using the SRA and IME together.

From Fig. 8, it can be observed that the IME module maintains high localization accuracy, regardless of the use of the SRA. In Fig. 8a, it is shown that by jointly using the SRA and IME modules, the aerial images captured by the UAV during the landing phase can also be used for localization. This effectively addresses the visual localization problem encountered in Experiment 1, where the IME module alone could not cope with significant changes in the altitude of the UAV.

Table 3 further demonstrates that integrating SRA with IME substantially improves the recall rate compared to using IME with fixed pixel sizes. This improvement stems from the SRA module dynamically resizing aerial images to align their spatial resolution closely with satellite images, thereby enhancing retrieval and local feature matching performance. Moreover, without SRA, the recall rate remains relatively high at location 1 when the fixed pixel size is set to 950 (closely matching the SRA-adjusted size of 971) and at location 2 with a fixed size of 1350 (near the SRA's online-adjusted 1340). Regardless of whether the SRA is

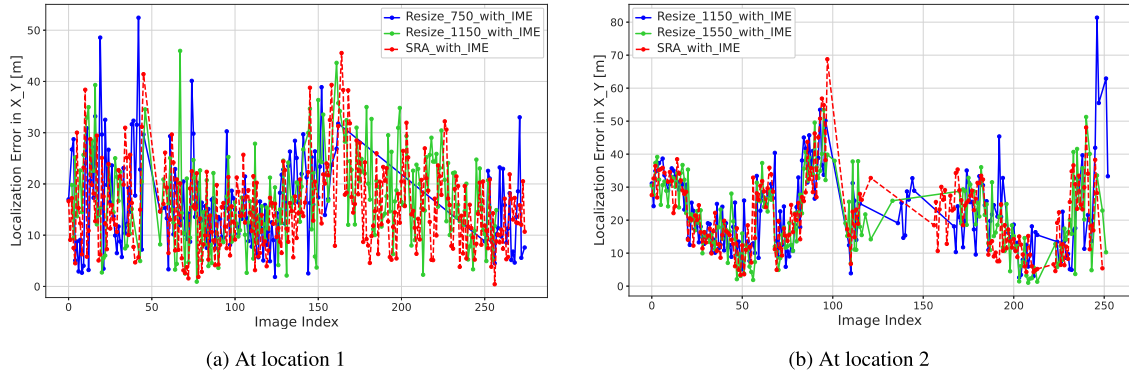


Fig. 8. Comparison of UAV localization errors at locations 1 and 2 when using the IME module alone versus the combined SRA and IME modules.

Table 3

A quantitative comparison of UAV localization errors at locations 1 and 2 when using the IME module alone versus the combined SRA and IME modules.

Location	Frames	Method	Recall (%)	Localization MAE[m]			Localization RMSE[m]		
				x	y	xy	x	y	xy
Location 1	275	Resize_750_with_IME	59.64	9.52	11.39	16.27	12.09	14.15	18.61
		Resize_950_with_IME	90.18	9.97	11.50	16.60	12.53	14.55	19.20
		Resize_1150_with_IME	77.45	9.48	12.11	17.15	11.96	15.05	19.22
		Resize_1350_with_IME	21.09	12.05	8.68	16.24	14.70	12.95	19.59
		SRA_with_IME	91.64	9.08	10.54	15.30	11.35	13.20	17.41
Location 2	274	Resize_950_with_IME	59.49	10.58	15.42	20.78	14.83	18.20	23.48
		Resize_1150_with_IME	63.87	10.75	17.08	22.19	14.99	20.44	25.35
		Resize_1350_with_IME	64.96	10.80	15.73	20.98	15.10	18.42	23.82
		Resize_1550_with_IME	63.14	10.84	14.97	20.67	15.80	17.19	23.34
		SRA_with_IME	68.61	10.57	16.98	21.70	14.95	19.41	24.50

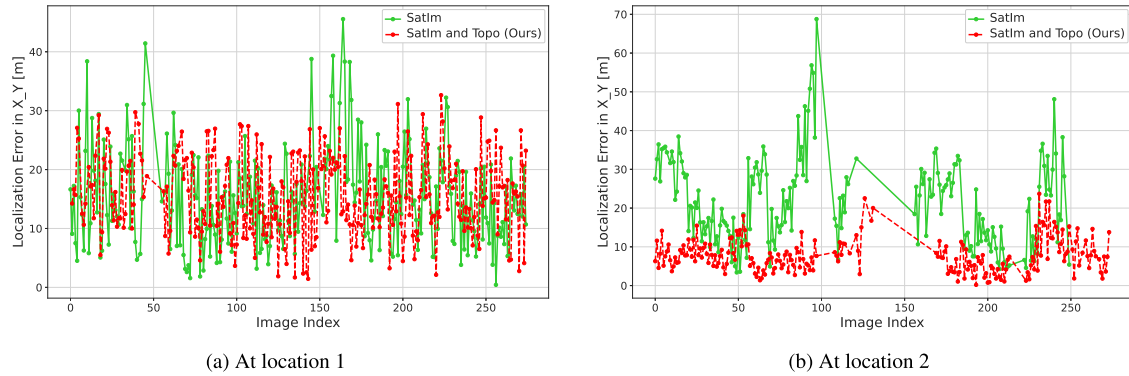


Fig. 9. Impact of two visual localization methods on the UAV overall localization errors along the x and y axes at locations 1 and 2.

used, the IME ensures that the UAV's localization accuracy varies within a reasonable range.

4.3.3. Experiment 3

This experiment aimed to validate that the visual localization method integrating satellite remote sensing imagery with terrain elevation imagery significantly outperforms the visual localization method relying solely on satellite imagery. The SRA and IME modules were employed to determine the available localization of aerial images, with the IME module's parameter settings consistent with those in Experiment 1. When the above two visual localization methods are applied at location 1 and location 2, the UAV's combined localization errors along the x and y axes are shown in Fig. 9, while the localization error along the z-axis is depicted in Fig. 10. The numerical results of the UAV localization errors are shown in the Table 4. "Satlm" refers to the UAV localization error estimated through the visual method relying solely on satellite remote sensing imagery; "Satlm and Topo (Ours)" refers to the localization er-

ror estimated through the visual method that integrates both satellite remote sensing imagery and terrain elevation.

From Fig. 9a and b reveal that, at location 1, the visual localization method combining satellite remote sensing imagery with terrain elevation imagery does not significantly change the UAV's comprehensive error along the x and y axes compared to using satellite imagery alone. Conversely, at location 2, the estimated comprehensive error on the x and y axes is significantly reduced. This is primarily because, when the camera adopts a nadir view for aerial photography, the elevation of the feature points has a minimal impact on their representation in the aerial images, only affecting the proportion of the feature points in the images. In contrast, when the camera uses an oblique view, the height of geographic feature points has a much greater influence on their pixel positions in aerial images.

From Fig. 10a and b, it is evident that the combined visual localization method, which integrates satellite remote sensing imagery with terrain elevation data, substantially reduces the UAV's localization error along the z-axis at both location 1 and location 2, compared to using

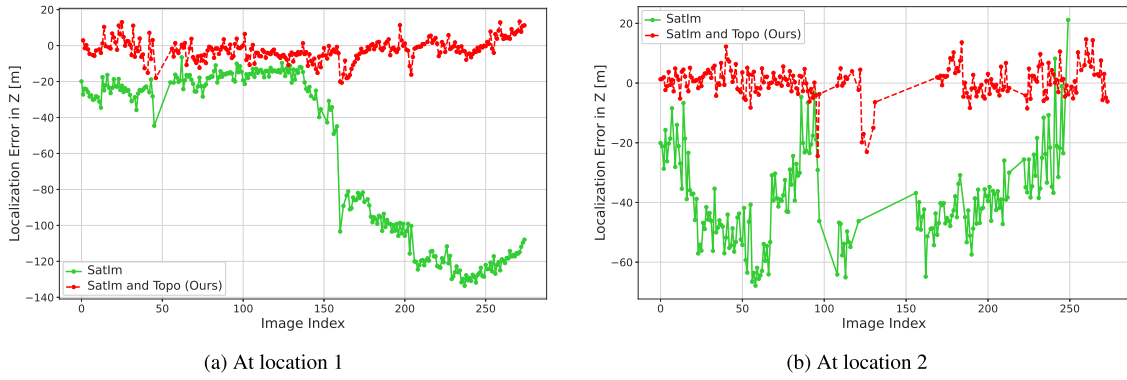


Fig. 10. Impact of two visual localization methods on the UAV z-axis localization errors at locations 1 and 2.

Table 4

Impact of two visual localization methods on the UAV localization errors at locations 1 and 2.

Location	Frames	Method	Recall (%)	Localization MAE[m]					Localization RMSE[m]				
				x	y	z	xy	xyz	x	y	z	xy	xyz
Location 1	275	Satlm	91.64	9.08	10.54	60.73	15.30	64.57	11.35	13.20	76.29	17.41	78.26
		Satlm and Topo	93.09	9.36	10.45	4.86	15.56	16.90	11.61	12.37	6.34	17.00	18.25
Location 2	274	Satlm	68.61	10.57	16.98	39.31	21.70	47.15	14.95	19.41	41.78	24.50	48.43
		Satlm and Topo	74.82	3.71	6.10	3.85	7.78	9.26	5.10	7.34	5.37	8.94	10.43

satellite imagery alone. Further examination of Table 4 reveals that at location 1, when the camera captures images with a nadir view, the estimated localization error on the z-axis reduces significantly from 60.73 m to 4.86 m. Additionally, the overall error across the x, y, and z axes drops from 64.57 m to 16.90 m. Simultaneously, at location 2, the localization error in the z-axis decreases from 39.31 m to 3.85 m, and the overall error decreases from 47.15 m to 9.26 m. These results demonstrate that incorporating terrain elevation information significantly enhances the UAV's localization accuracy, achieving geographical localization in three degrees of freedom. Additionally, comparing the nadir view at location 1 with the oblique view at location 2, we observe that the nadir view yields a higher recall rate. This is because the nadir view images have a smaller perspective difference from the satellite remote sensing imagery, facilitating easier retrieval from the satellite image database.

4.3.4. Experiment 4

This experiment was conducted to visually demonstrate the superiority of our method by comparing the flight heights and trajectories measured with RTK as a reference to the flight heights and trajectories derived by our method. It's important to note that the altitude derived from the visual geo-localization method using only satellite remote sensing imagery is a relative altitude from the takeoff point, while the altitude obtained by combining satellite imagery with terrain elevation data is the ellipsoidal height, consistent with the reference coordinate system used by satellite navigation system elevation positioning results. To facilitate comparison, the relative altitude from the first method is converted to absolute altitude by adding the elevation of the takeoff point. The UAV's flight trajectory on satellite imagery is illustrated in Fig. 11, and its flight altitude in the geographic coordinate system is shown in Fig. 12. "Reference Truth" refers to the localization results measured by RTK; "Satlm" refers to the localization results achieved by the visual localization method relying solely on satellite remote sensing imagery; and "Satlm and Topo (Ours)" refers to the localization results from our proposed method that integrates both satellite remote sensing imagery and terrain elevation data.

From Fig. 11a and b, it is evident that the UAV flight trajectories estimated by the two visual localization methods is very close, and also closely aligns with the localization obtained by RTK, demonstrating their effectiveness in inferring the UAV's position. Further analysis of

Fig. 12a and b reveals that the altitude accuracy achieved by the visual localization method integrating satellite remote sensing imagery with terrain elevation imagery is markedly superior to that achieved by the method relying solely on satellite remote sensing imagery. Therefore, by combining satellite remote sensing imagery with terrain elevation imagery, we have extended the traditional two-dimensional planar localization in latitude and longitude directions to three-dimensional spatial localization.

Additionally, the visual localization method that combines satellite remote sensing imagery and terrain elevation imagery takes terrain factors into account, it can be applied to areas with complex terrain such as hills, which fully demonstrates the superiority of this method. Furthermore, the flight altitude derived from this method is an absolute altitude, which will aid in the precise navigation of the UAV in complex terrains, thereby expanding the potential applications of the UAV.

4.3.5. Experiment 5

This is to verify the rationality of setting the relative errors of invariant moments h_0 and h_1 and the area relative error when aerial images and satellite images have good feature matches. The invariant moments and area relative errors of aerial images with reliable feature matches under different camera viewing angles are shown in Fig. 13. The horizontal axis denotes the image index, and the vertical axis denotes the relative error. The blue dashed line indicates the relative error of invariant moment h_0 , the green dashed line indicates the relative error of invariant moment h_1 , and the red dashed line indicates the area relative error.

From Fig. 13a, it can be seen that when the camera captures images with a nadir angle, both the relative errors of the invariant moments and the area relative error are close to 0, with the maximum absolute value of the relative error being approximately 0.12. This is because when the camera captures images from a nadir view, the shape and size of the aerial images remain basically unchanged before and after projection. In Fig. 13b, the relative error range of invariant moment h_0 is approximately $[-0.6, -0.2]$, the relative error range of invariant moment h_1 is approximately $[-0.4, -0.2]$, and the area relative error ranges from about $[0.2, 0.5]$. It can be observed that compared to the nadir view, when the camera captures images from an oblique view, the absolute values of the relative errors of the invariant moments and the area rela-

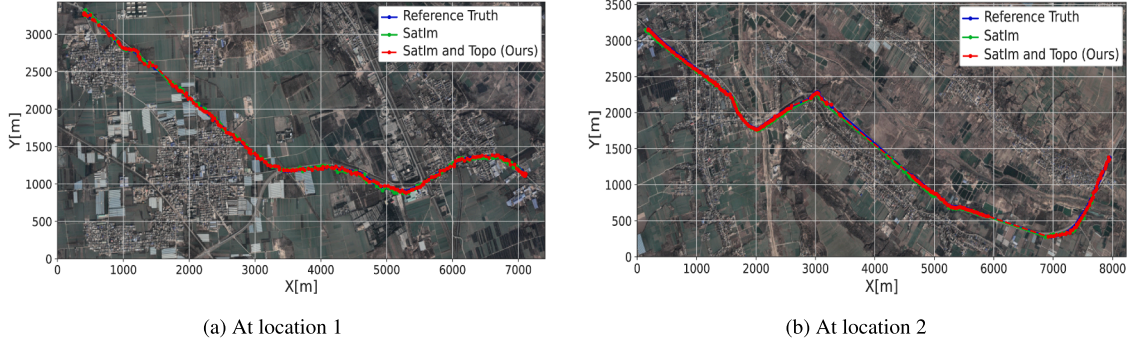


Fig. 11. Comparison between the flight trajectories estimated by our two proposed methods and the flight trajectory obtained through RTK.

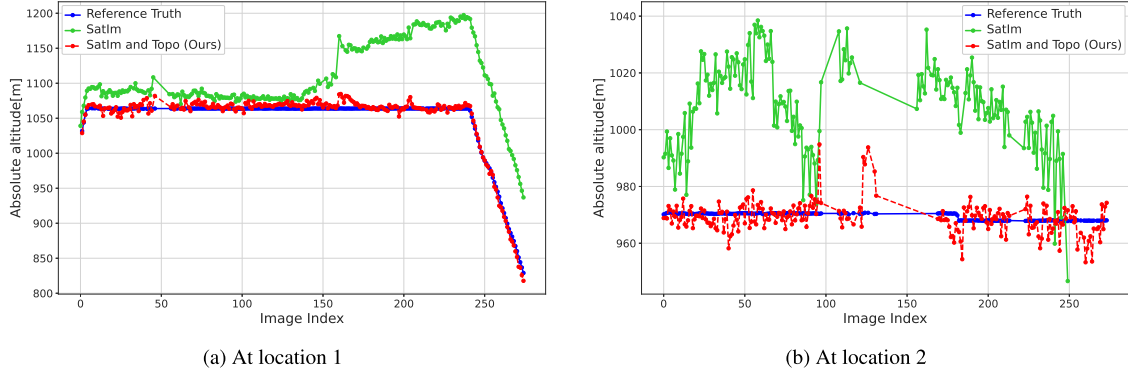


Fig. 12. Comparison between the flight altitudes estimated by our two proposed methods and the flight heights obtained through RTK.

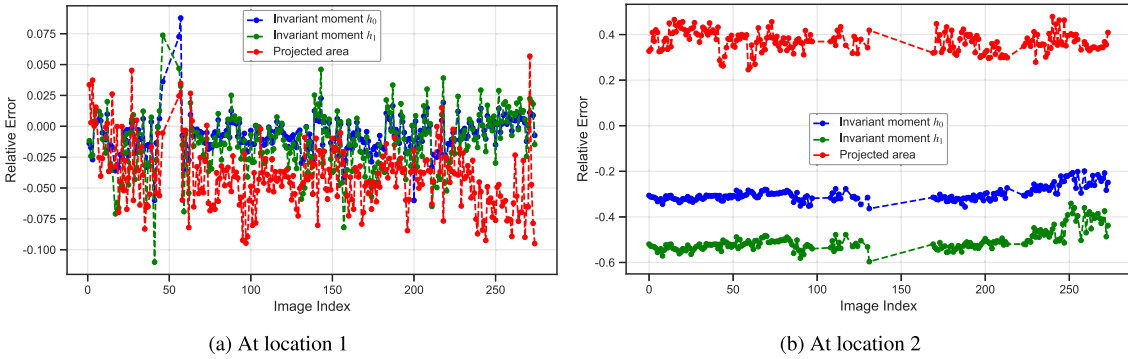


Fig. 13. Invariant moment relative error and area relative error of the camera under different viewing angle conditions.

tive error increase to some extent. However, when there is robust feature matching between the aerial and satellite images, the relative errors of the invariant moments and area in the local central region of the aerial image projected onto the satellite image do not change significantly. Thus, under the condition that the area relative error is within $[-0.3, 1]$, if the forward tilt angle $\tau \in [-5^\circ, 5^\circ]$ and $\delta_{h_0}, \delta_{h_1} \in [-0.2, 0.2]$, or if the forward tilt angle $\tau \notin [-5^\circ, 5^\circ]$ and $\delta_{h_0}, \delta_{h_1}$ are not within $[-0.1, 0.1]$ but within $[-4.5, 4.5]$, the aerial image is considered to have good local feature matching performance.

To demonstrate the superiority and potential for enhancement of our proposed method, we maintained the global feature extraction network OPIBL as is, while substituting the network SuperPoint with SIFT (Li et al., 2024b), ALIKED (Guo et al., 2024), and DISK (Zumaya et al., 2024), proposing three new visual methods: “OPIBL + SIFT + LG (LightGlue)”, “OPIBL + ALIKED + LG”, and “OPIBL + DISK + LG”. Additionally, we still use SuperPoint to extract local features and “VGG16 + NetVLAD + PCA” as the global feature extraction network, which was trained on the Tokyo dataset (Torii et al., 2018), proposing the

visual method “VGG16 + SP + LG”. Meanwhile, we compared our proposed methods with the visual method presented by Chen et al. (2021), which employs “MobileNetV2 + NetVLAD + PCA” as the global feature extraction network and uses SuperPoint and SuperGlue for local feature extraction and matching. The aerial image recall and localization errors obtained by these methods are summarized in Table 5.

In Table 5, our proposed method is compared with “VGG16 + SP + LG” and the method “MobileNetV2 + SP + SG” presented by Chen et al. (2021). The results show that the choice of the global feature extraction network significantly affects the recall rate and localization accuracy. This is because different global feature networks generate distinct global image descriptors for the same input image, which may result in different outcomes when searching for satellite images that match the aerial image’s global descriptor within the satellite image database. Consequently, different global feature extraction networks may cause the aerial image to match local features with different satellite images, thereby impacting the recall rate and localization accuracy. For example, compared to “VGG16 + SP + LG”, our proposed method increased

Table 5

Comparison of localization errors of several visual localization methods with our proposed method at location 1 and 2.

Location	Method	Recall (%)	Localization MAE(m)					Localization RMSE(m)				
			x	y	z	xy	xyz	x	y	z	xy	xyz
location 1	OPIBL + Aliked + LG	82.55	11.84	13.1	4.8	19.47	20.5	15.21	15.87	6.09	21.98	22.81
	OPIBL + Disk + LG	61.82	11.21	12.14	4.8	18.42	19.49	14.2	15.46	6.31	20.99	21.92
	OPIBL + Sift + LG	57.82	14.2	17.54	5.72	24.73	25.94	17.31	21.58	7.5	27.66	28.66
	VGG16 + SP + LG	77.09	9.55	12.58	5.12	17.59	18.89	12.54	16.51	6.82	20.73	21.82
	Chen et al. (2021)	66.88	10.65	11.82	5.78	17.55	19.09	13.25	15.07	7.24	20.06	21.33
	Ours	93.09	9.36	10.45	4.86	15.56	16.90	11.61	12.37	6.34	17.00	18.25
location 2	OPIBL + Aliked + LG	57.66	3.98	5.14	5.00	7.20	9.51	5.42	6.51	7.07	8.47	11.04
	OPIBL + Disk + LG	10.95	4.78	5.76	4.91	8.12	10.08	6.22	6.87	6.47	9.06	11.23
	OPIBL + Sift + LG	/	/	/	/	/	/	/	/	/	/	/
	VGG16 + SP + LG	36.5	4.98	6.54	4.5	8.94	10.57	6.91	7.67	5.9	10.32	11.89
	Chen et al. (2021)	33.03	3.73	6.56	5.85	8.21	10.78	5.47	8.07	7.23	9.75	12.14
	Ours	74.82	3.71	6.1	3.85	7.78	9.26	5.1	7.34	5.37	8.94	10.43

the recall rate at location 1 from 77.09% to 93.09%, and simultaneously reduced the overall error across the x, y, and z axes from 18.89 m to 16.90 m. At location 2, the recall rate improved significantly from 36.50% to 74.82%, and the overall error decreased from 10.57 m to 9.26 m.

Comparing the localization methods “OPIBL + SIFT + LG,” “OPIBL + ALIKED + LG,” and “OPIBL + DISK + LG” with our proposed localization method, it can be seen that the choice of local feature extraction network significantly influences recall rates and localization accuracy. At Location 1 (near-nadir view), our method achieves a recall rate of 93.09%, outperforming “OPIBL + ALIKED + LG”(82.55%), “OPIBL + DISK + LG”(61.82%), and “OPIBL + SIFT + LG”(57.82%). Regarding localization error (xyz mean absolute error, MAE), our method yields an error of only 16.90 m, significantly outperforming “OPIBL + SIFT + LG” with 25.94 m, as well as “OPIBL + ALIKED + LG” with 20.5 m and “OPIBL + DISK + LG” with 19.49 m. At Location 2 (oblique view), the performance gap is even more pronounced. Our method maintains a recall rate of 74.82% for aerial images, whereas “OPIBL + ALIKED + LG” drops to 57.66%, “OPIBL + DISK + LG” sharply declines to 10.95%, and “OPIBL + SIFT + LG” fails to produce effective localization (denoted as “/” in the table). Additionally, our proposed method also achieves better localization accuracy compared to “OPIBL + ALIKED + LG” (9.51 m) and “OPIBL + DISK + LG” (10.08 m). These results indicate that even when the same global feature extraction network retrieves the actual satellite images, differences in local feature descriptors extracted by different local extraction networks can hinder effective local feature matching, thereby affecting recall rate and localization accuracy. Thus, selecting an appropriate feature extraction network is crucial for improving both recall rate and localization accuracy. Additionally, the IME module effectively filters out aerial images that are unsuitable for localization, ensuring that localization accuracy fluctuates within a reasonable range. In the future, with the advancement of deep learning, the use of more advanced neural networks to replace the existing networks is expected to further improve the localization precision of the UAV.

5. Conclusions

The primary contribution of this work is the design and implementation of a practical, system-level framework for UAV visual geo-localization under conditions where satellite navigation is denied and accurate prior information and image data cannot be obtained through pre-flight. This framework innovatively integrates: (1) deep neural networks for feature representation (OpenIBL, SuperPoint); (2) novel algorithmic modules supporting resolution-adaptive processing and matching verification (SRA, IME); and (3) multi-source geospatial data, including satellite imagery and terrain elevation imagery. Through the synergistic integration of these technologies, the system extends traditional two-dimensional (latitude-longitude) localization to full three-dimensional spatial localization, requiring no prior information about

the takeoff point, and possesses efficient re-localization capabilities after prolonged GPS failure.

We established a satellite image database covering an area of 26.9 km in width and 7.03 km in height, conducting experimental validation at two locations: location 1 with a nadir view and location 2 with an oblique view at approximately 45°. The results demonstrate that the integration of the SRA and IME modules effectively filters out aerial images with substantial localization errors, significantly enhancing UAV localization accuracy. Moreover, during takeoff and landing phases, where flight altitude varies considerably, the method maintains high-precision visual localization. Further incorporating terrain elevation can effectively boost localization performance. Compared with the method relying solely on satellite imagery, our terrain elevation-fused method achieves apparent performance improvements: at location 1, the recall rate improved from 91.64% to 93.09%, and the overall error was reduced from 64.57 m to 16.90 m. Simultaneously, at location 2, the recall rate increased from 68.61% to 74.82%, and the overall error dropped from 47.15 m to 9.26 m. These results not only verify the effectiveness of the proposed method but also specifically address the core limitations identified in existing research. First, regarding the reliance on pre-built 3D terrain models, our method eliminates the need for pre-flight 3D model construction and achieves localization solely based on satellite images and terrain elevation data. The low localization errors in unmapped areas (16.90 m in Location 1 and 9.26 m in Location 2) directly demonstrate its applicability to regions without prior image data. Second, concerning the high computational overhead in large-scale scenes, the matching effectiveness evaluation strategy of the SRA and IME modules reduces redundant computations by filtering irrelevant images, effectively alleviating computational burden. Third, addressing the lack of support for non-strict nadir photography, the 74.82% recall rate in the 45° oblique view scenario overcomes the viewing angle limitations of traditional methods, enabling robust localization under non-nadir angles.

Despite these improvements, our method has inherent limitations that warrant attention. First, it relies on pre-built satellite image and terrain elevation databases, leading to significant positioning degradation in scenarios where database timeliness is compromised (e.g., post-disaster terrain changes). Second, while terrain elevation data is used to provide elevation information for matched feature points, the current framework overlooks the elevation details of ground objects (e.g., buildings, vegetation), limiting the refinement of 3D coordinates. Third, the core work of this study focuses on the cascade integration and experimental validation of the visual localization system. Although the impact of various advanced neural network models on localization performance has been analyzed, there remains a lack of in-depth theoretical analysis of feature representation mechanisms and network architectures, and the real-time operational frequency requirements for deploying the algorithm on UAV embedded systems have not been fully considered. A thorough understanding of the internal workings of neural networks is precisely the key prerequisite for achieving lightweight network design.

To address these limitations, future work will focus on three directions: (1) leveraging real-time image stitching and sparse point cloud reconstruction technologies to achieve online mapping of dynamically changing areas, thereby replacing outdated satellite imagery and terrain elevation databases; simultaneously providing more refined geographic information support for matched feature points to further enhance visual localization accuracy. (2) Integrating multi-sensor data to enhance system reliability—using IMUs to provide high-frequency pose estimates to compensate for low-frequency output and potential interruptions in visual localization, while incorporating altimeters to further enhance localization accuracy. (3) Conducting in-depth research on the internal mechanisms of neural networks, reducing matching complexity through feature pruning and descriptor compression, achieving feature tracking and reuse based on motion continuity between adjacent frames, and deploying lightweight neural network architectures to increase single-frame processing speed to meet the frame rate requirements for real-time UAV operation. These refinements aim to develop a more robust and accurate UAV visual geo-localization system, serving as a reliable alternative to satellite navigation in GNSS-denied environments.

CRedit authorship contribution statement

Xiong Qiu: Writing – review & editing, Writing – original draft, Software; **Shouyi Liao:** Methodology, Conceptualization; **Dongfang Yang:** Investigation, Data curation; **Ting Li:** Software, Investigation; **Shicheng Wang:** Software, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Al-Qaysi, Z. T., Salih, M., Shuwandy, M., Ahmed, M. A., & Altarazi, Y. (2023). Multi-tiered CNN model for motor imagery analysis: Enhancing UAV control in smart city infrastructure for industry 5.0. *Applied Data Science and Analysis*, 2023, 88–101.
- Albdair, M., mohsin, Z. R., Saihood, A. A., Hamad, A. M., & Sahi, A. (2025). Secured multi-objective optimisation-based protocol for reliable data transmission in underwater wireless sensor networks. *Mesopotamian Journal of CyberSecurity*, 5(1), 216–239.
- Alkhatib, M., Nayfeh, M., Al Shamaileh, K., Kaabouch, N., & Devabhaktuni, V. (2025). A return-to-home unmanned aerial vehicle navigation solution in global positioning system denied environments via bidirectional long short-term memory reverse flightpath prediction. *Engineering Applications of Artificial Intelligence*, 140, 109729.
- Aslan, M. F., Durdu, A., Yusefi, A., & Yilmaz, A. (2022). Hvionet: A deep learning based hybrid visual-inertial odometry approach for unmanned aerial system position estimation. *Neural Networks*, 155, 461–474.
- Chen, L., Jin, Z., Shao, K., Wang, G., He, S., Stojanović, V., Bahri, P. A., & Wang, H. (2024). Adaptive integral terminal sliding mode control of unmanned bicycle via ELM and barrier function. *Robotica*, 42, 2635–2657.
- Chen, S., Wu, X., Mueller, M. W., & Sreenath, K. (2021). Real-time geo-localization using satellite imagery and topography for unmanned aerial vehicles. In *2021 IEEE/RSJ international conference on intelligent robots and systems (IROS)* (pp. 2275–2281).
- Couturier, A., & Akhloufi, M. A. (2021). A review on absolute visual localization for UAV. *Robotics and Autonomous Systems*, 135, 103666.
- DeTone, D., Malisiewicz, T., & Rabinovich, A. (2018). Superpoint: Self-supervised interest point detection and description. In *2018 IEEE/CVF conference on computer vision and pattern recognition workshops (CVPRW)* (pp. 337–33712).
- Faber, K., Corizzo, R., Sniezynski, B., & Japkowicz, N. (2023). Vlad: Task-agnostic vae-based lifelong anomaly detection. *Neural Networks*, 165, 248–273.
- Fan, X., Liu, S., Liu, S., Zhao, L., & Li, R. (2025a). Aapmatcher: Adaptive attention pruning matcher for accurate local feature matching. *Neural Networks*, 188, 107403.
- Fan, Z., Zhang, L., Wang, X., Shen, Y., & Deng, F. (2025b). LiDAR, IMU, and camera fusion for simultaneous localization and mapping: a systematic review. *Artificial Intelligence Review*, 58(6), 174.
- Goforth, H., & Lucey, S. (2019). Gps-denied UAV localization using pre-existing satellite imagery. *2019 IEEE international conference on robotics and automation*, (pp. 2974–2980).
- Goh, H., Thome, N., Cord, M., & Lim, J.-H. (2014). Learning deep hierarchical visual feature coding. *IEEE Transactions on Neural Networks and Learning Systems*, 25(12), 2212–2225.
- Guo, D., Wu, S., Weng, D., Gao, C., & Li, W. (2024). Invariant feature matching in space-craft rendezvous and docking optical imaging based on deep learning. *Remote Sensing*, 16(24), 4690.
- Gurgu, M.-M., Queralta, J. P., & Westerlund, T. (2022). Vision-based GNSS-free localization for UAVs in the wild. In *2022 7th international conference on mechanical engineering and robotics research (ICMERR)* (pp. 7–12).
- Gürgünoğlu, D., Dulek, B., & Gezici, S. (2022). Power adaptation for vector parameter estimation according to fisher information based optimality criteria. *Signal Processing*, 192, 108390.
- Khaliq, A., Milford, M., & Garg, S. (2022). Multires-netVLAD: Augmenting place recognition training with low-resolution imagery. *IEEE Robotics and Automation Letters*, 7(2), 3882–3889.
- Le Jeune, P., Bahaduri, B., & Mokraoui, A. (2025). A comparative attention framework for better few-shot object detection on aerial images. *Pattern Recognition*, 161, 111243.
- Li, H., Deuser, F., Yin, W., Luo, X., Walther, P., Mai, G., Huang, W., & Werner, M. (2025a). Cross-view geolocalization and disaster mapping with street-view and VHR satellite imagery: A case study of hurricane IAN. *ISPRS Journal of Photogrammetry and Remote Sensing*, 220, 841–854.
- Li, P., Xu, F., Wang, J., Guo, H., Miao, L., & Su, M. (2024a). C-LFNet: Central-local feature 3D point cloud instance segmentation network for robot bin-picking. *Measurement*, 225, 113916.
- Li, X., Zhu, G., Wang, S., Zhou, Y., & Zhang, X. (2024b). Deep reverse attack on SIFT features with a coarse-to-fine GAN model. *IEEE Transactions on Circuits and Systems for Video Technology*, 34(7), 6391–6402.
- Li, Y., Xia, J., Liao, H., Lan, X., & Jiang, W. (2025b). The SIFT based two-stage STC decoupled learning method for long-tailed SAR target recognition. *Neurocomputing*, 633, 129747.
- Lin, Z., Zhang, Q., Tian, Z., Yu, P., Ye, Z., Zhuang, H., & Lan, J. (2025). Slam2: Simultaneous localization and multimode mapping for indoor dynamic environments. *Pattern Recognition*, 158, 111054.
- Lindenberger, P., Sarlin, P.-E., & Pollefeys, M. (2023). Lightglue: Local feature matching at light speed. In *2023 IEEE/CVF international conference on computer vision (ICCV)* (pp. 17581–17592).
- Liu, T., Zhang, B., Tan, Q., Zhou, J., Yu, S., Zhu, Q., & Bian, Y. (2024). Immersive human-machine teleoperation framework for precision agriculture: Integrating UAV-based digital mapping and virtual reality control. *Computers and Electronics in Agriculture*, 226, 109444.
- Liu, X., Zhang, S., Huang, T., & Tian, Q. (2020). E2brows: An end-to-end bag-of-words model via deep convolutional neural network for image retrieval. *Neurocomputing*, 395, 188–198.
- Luo, X., Wan, X., Gao, Y., Tian, Y., Zhang, W., & Shu, L. (2024). Jointloc: A real-time visual localization framework for planetary UAVs based on joint relative and absolute pose estimation. In *2024 IEEE/RSJ international conference on intelligent robots and systems (IROS)* (pp. 3348–3355).
- Lynen, S., Zeisl, B., Aiger, D., Bosse, M., Hesch, J., Pollefeys, M., Siegwart, R., & Sattler, T. (2020). Large-scale, real-time visual-inertial localization revisited. *International Journal of Robotics Research*, 39(9), 1061–1084.
- Nafea, A., Alameri, S., Majeed, R. R., Khalaf, A., & Al-Ani, M. M. (2024). A short review on supervised machine learning and deep learning techniques in computer vision. *Babylonian Journal of Machine Learning*, 2024, 48–55.
- Patel, B., Barfoot, T. D., & Schoellig, A. P. (2020). Visual localization with google earth images for robust global pose estimation of UAVs. In *2020 IEEE international conference on robotics and automation (ICRA)* (pp. 6491–6497).
- Qiu, X., Liao, S., Yang, D., Li, Y., & Wang, S. (2025). High-precision visual geo-localization of UAV based on hierarchical localization. *Expert Systems with Applications*, 267, 126064.
- Qiu, X., Yang, D., Liao, S., Wang, S., & Li, Y. (2024). Image moment extraction based aerial photo selection for UAV high-precision geolocation without GPS. *Measurement*, 226, 114141.
- Si, L., Wang, Z., Li, J., Wei, D., & Zhao, S. (2023). A novel positioning method of anti-punching drilling robot based on the fusion of multi-IMUs and visual image. *ISA Transactions*, 137, 730–746.
- Song, S., Jiang, Y., Song, X., & Stojanovic, V. (2025). Composite neural learning-based adaptive actuator failure compensation control for full-state constrained autonomous surface vehicle. *Neural Computing and Applications*, 37(8), 6369–6381.
- Torii, A., Arandjelović, R., Sivic, J., Okutomi, M., & Pajdla, T. (2018). 24/7 place recognition by view synthesis. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 40(2), 257–271.
- Wang, J., & Qi, Y. (2024). Multi-level feature fusion and joint refinement for simultaneous object pose estimation and camera localization. *Neural Networks*, 174, 106238.
- Wang, R., Shi, Y., & Cao, W. (2019). Ga-surf: A new speeded-up robust feature extraction algorithm for multispectral images based on geometric algebra. *Pattern Recognition Letters*, 127, 11–17.
- Wang, T., Zheng, Z., Sun, Y., Yan, C., Yang, Y., & Chua, T.-S. (2024a). Multiple-environment self-adaptive network for aerial-view geo-localization. *Pattern Recognition*, 152, 110363.
- Wang, W., Min, H., Wu, X., Yang, L., Yan, C., Fang, Y., & Zhao, X. (2024b). LGD: A fast place recognition method based on the fusion of local and global descriptors. *Expert Systems with Applications*, 251(1), 123996.
- Wilks, C., Ahmed, O., Baker, D. N., Zhang, D., Collado-Torres, L., & Langmead, B. (2021). Megadept: efficient coverage quantification for bigwigs and BAMs. *Bioinformatics*, 37(18), 3014–3016.
- Wu, R., Cheng, X., Zhu, J., Liu, Y., Zhang, M., & Yan, S. (2024). Uavd4l: A large-scale dataset for UAV 6-DoF localization. In *2024 international conference on 3D vision (3DV)* (pp. 1574–1583).
- Yao, F., Lan, C., Wang, L., Wan, H., Gao, T., & Wei, Z. (2024). GNSS-denied geolocalization of UAVs using terrain-weighted constraint optimization. *International Journal of Applied Earth Observation and Geoinformation*, 135, 104277.

- Yass, W., & Faris, M. (2023). Integrating computer vision, web systems and embedded systems to develop an intelligent monitoring system for violating vehicles. *Babylonian Journal of Internet of Things*, 2023, 69–73.
- Ye, Q., Luo, J., & Lin, Y. (2024). A coarse-to-fine visual geo-localization method for GNSS-denied UAV with oblique-view imagery. *ISPRS Journal of Photogrammetry and Remote Sensing*, 212, 306–322.
- Yuan, Y., Sun, B., & Liu, G. (2022). Dual attention and dual fusion: An accurate way of image-based geo-localization. *Neurocomputing*, 500, 965–977.
- Zhang, B.-J., Liu, G.-H., Li, Z., & Song, S.-X. (2024). Image retrieval using compact deep semantic correlation descriptors. *Information Processing and Management*, 61(3), 103608.
- Zhang, Y., Ji, Z., Pang, Y., & Han, J. (2025). Hierarchical and complementary experts transformer with momentum invariance for image-text retrieval. *Knowledge-Based Systems*, 309, 112912.
- Zumaya, C. R. C., Catalano, I., & Queralta, J. P. (2024). Building better models: Benchmarking feature extraction and matching for structure from motion at construction sites. *Remote Sensing*, 16(16), 2974.